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Designing a Web Generator for MSMEs for Business Digitalization

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***Abstract*—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam sed dapibus erat, nec varius nulla. Ut vitae nunc lorem. Etiam non ornare velit, sed rutrum odio. Vivamus a lacinia velit. Integer fringilla tincidunt rhoncus. Etiam hendrerit pharetra dapibus. Curabitur in luctus enim. Aenean lacus lectus, imperdiet ut lectus sit amet, mollis convallis enim. Ut tempor elit laoreet nulla tempus iaculis. Aliquam erat volutpat. Aliquam nec turpis ut metus venenatis dapibus et in justo.**

***Index Terms*—keyword1, keyword2, keyword3**

I. Introduction

The Indonesian digital economy is currently experiencing a period of rapid growth, with projections estimating that its value will reach USD 150 billion by 2025. Despite this potential, the Micro, Small, and Medium Enterprises (MSME) sector—which serves as the backbone of the national economy by contributing 61% to the GDP—continues to face significant challenges in fully participating in this digital transformation. Research by Rajagukguk and Rusadi (2025) highlights a clear "adoption gap," revealing that while 65% of MSMEs recognize the necessity of digital technology, only 23% are able to utilize it consistently.

This disparity stems largely from the incompatibility of existing technological tools with the needs and capabilities of non-technical MSME owners. Currently, these business owners are often forced into difficult trade-offs: either adopting marketplace platforms that impose administrative fees—averaging 6.5%—which erode thin profit margins and limit control over customer data, or attempting to build independent websites using complex Content Management Systems (CMS) like WordPress, which often prove too technically demanding for those with limited digital literacy. Furthermore, popular international website builders (e.g., Wix) frequently lack native integration with local Indonesian digital habits, such as QRIS payments and automated WhatsApp communication.

To address these barriers, this research proposes the development of a low-code/no-code web generator platform specifically tailored for Indonesian MSMEs. This platform aims to provide a "middle ground" solution that is intuitive for non-technical users, cost-effective by eliminating commission-based dependencies, and locally relevant through built-in support for QRIS and WhatsApp. The objective of this study is to design a functional Minimum Viable Product (MVP) and empirically demonstrate, through System Usability Scale (SUS) benchmarking, that this platform offers a significantly higher level of usability and ease of creation for MSME owners compared to conventional CMS alternatives. By providing a professional yet accessible digital presence, this research contributes to the broader effort of democratizing digital technology for small businesses across Indonesia.

II. Literature Review

A. Digital Transformation of MSMEs

Micro, Small, and Medium Enterprises (MSMEs) represent a fundamental pillar of economic development, particularly in emerging economies. They contribute significantly to employment generation, income distribution, and national Gross Domestic Product (GDP) [9]. Despite their strategic importance, MSMEs often face structural challenges in adapting to the rapidly evolving digital economy.

The advancement of digital technologies has created new opportunities for MSMEs to expand market reach, improve operational efficiency, and enhance customer engagement. However, the adoption of digital tools among MSMEs remains uneven. Several studies have identified the existence of an adoption gap, where awareness of digital technologies does not necessarily translate into actual implementation [6].

According to OECD, digital transformation can significantly improve productivity and competitiveness among small businesses, but its success depends on accessibility, affordability, and usability of the technology [6]. In many cases, MSMEs lack the necessary digital skills, financial resources, and technical support to fully utilize available solutions.

Furthermore, Vial (2019) emphasizes that digital transformation is not merely a technological shift but also an organizational and cultural change [1]. For MSMEs, this transformation is particularly challenging due to their limited resources and informal business structures.

B. Limitations of Existing Digital Platforms

Existing digital solutions for MSMEs generally fall into two categories: marketplace platforms and Content Management Systems (CMS). Marketplace platforms offer ease of use but impose transaction fees and limit control over customer data [6].

On the other hand, CMS platforms such as WordPress provide greater flexibility but require technical expertise, making them less suitable for non-technical users. Website builders such as Wix attempt to simplify development; however, they are typically designed for global users and may not support local business practices effectively.

C. Low-Code/No-Code Development

Low-code and no-code development platforms have emerged as a promising approach to reduce technical barriers in application development. These platforms allow users to create applications through visual interfaces with minimal coding knowledge. Research by Sahay et al. shows that low-code platforms significantly accelerate development and enable non-programmers to build applications [4]. Similarly, Kumar and Pandey found that no-code platforms improve accessibility and reduce development time for small businesses [5]. However, most platforms remain generic and lack localization for specific business contexts.

D. User-Centered Design (UCD)

User-Centered Design (UCD) focuses on designing systems based on user needs, capabilities, and limitations. This approach is particularly important for MSMEs with low digital literacy. Studies show that applying UCD improves usability and user satisfaction by reducing system complexity and aligning design with real-world usage [7].

E. Usability Evaluation Using System Usability Scale (SUS)

Usability plays a critical role in system adoption. The System Usability Scale (SUS), introduced by Brooke, is a widely used tool for evaluating usability [2].

Bangor et al. further validated SUS as a reliable method for assessing usability across different systems, including web applications [3]. SUS allows quantitative comparison between systems, making it suitable for evaluating new platforms against existing solutions.

F. Research Gap

Based on the reviewed literature, several research gaps are identified:

- Lack of localized digital solutions tailored for MSMEs
- High technical complexity for non-expert users
- Limited adoption of user-centered design principles
- Insufficient empirical usability comparison using standardized metrics

G. Research Contribution

This study proposes a low-code/no-code web generator platform specifically designed for MSMEs. The platform integrates localized features and is developed using a user-centered approach.

III. METHOD

Jelaskan metode penelitian kamu.

IV. RESULTS AND DISCUSSION

Tampilkan hasil dan pembahasan.

V. CONCLUSION

Tulis kesimpulan.

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REFERENCES

[1] G. Vial, “Understanding digital transformation: A review and a research agenda,” Journal of Strategic Information Systems, vol. 28, no. 2, pp. 118–144, 2019.
[2] J. Brooke, “SUS: A ‘quick and dirty’ usability scale,” in Usability Evaluation in Industry. London, U.K.: Taylor & Francis, 1996.
[3] A. Bangor, P. Kortum, and J. Miller, “An empirical evaluation of the system usability scale,” International Journal of Human-Computer Interaction, vol. 24, no. 6, pp. 574–594, 2008.
[4] A. Sahay, C. Indamutsa, D. Di Ruscio, and I. Malavolta, “Supporting the understanding and comparison of low-code development platforms,” in Proc. ICSE, 2020.
[5] R. Kumar and A. Pandey, “Adoption of no-code development platforms,” International Journal of Computer Applications, vol. 183, no. 12, 2021.
[6] OECD, “The Digital Transformation of SMEs,” OECD Publishing, 2021.
[7] Y. Zhou, “User-centered design in digital systems,” Journal of UX Studies, vol. 5, no. 1, 2022.
[8] M. Armbrust et al., “A view of cloud computing,” Communications of the ACM, 2010.
[9] T. Beck and A. Demirgüç-Kunt, “SMEs, growth, and constraints,” Journal of Banking & Finance, 2006.